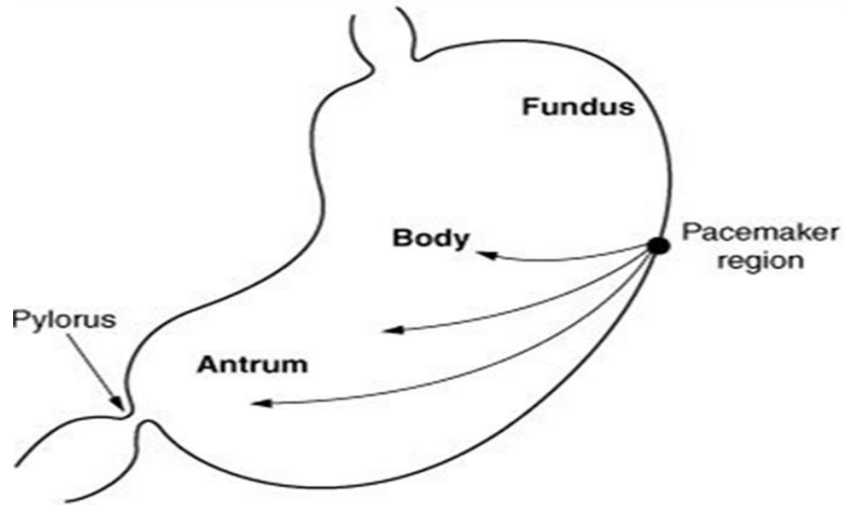


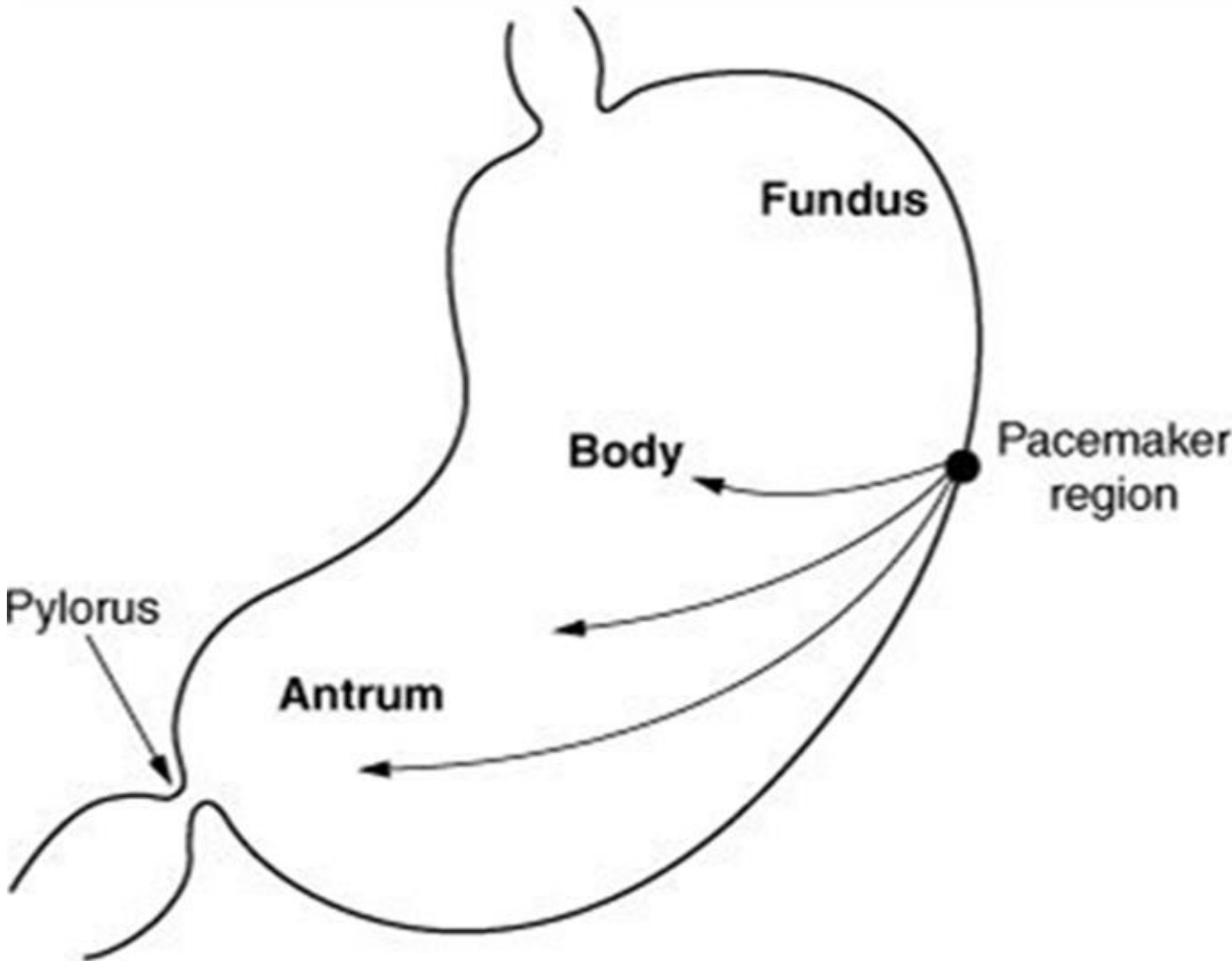
Physiology of the GIT

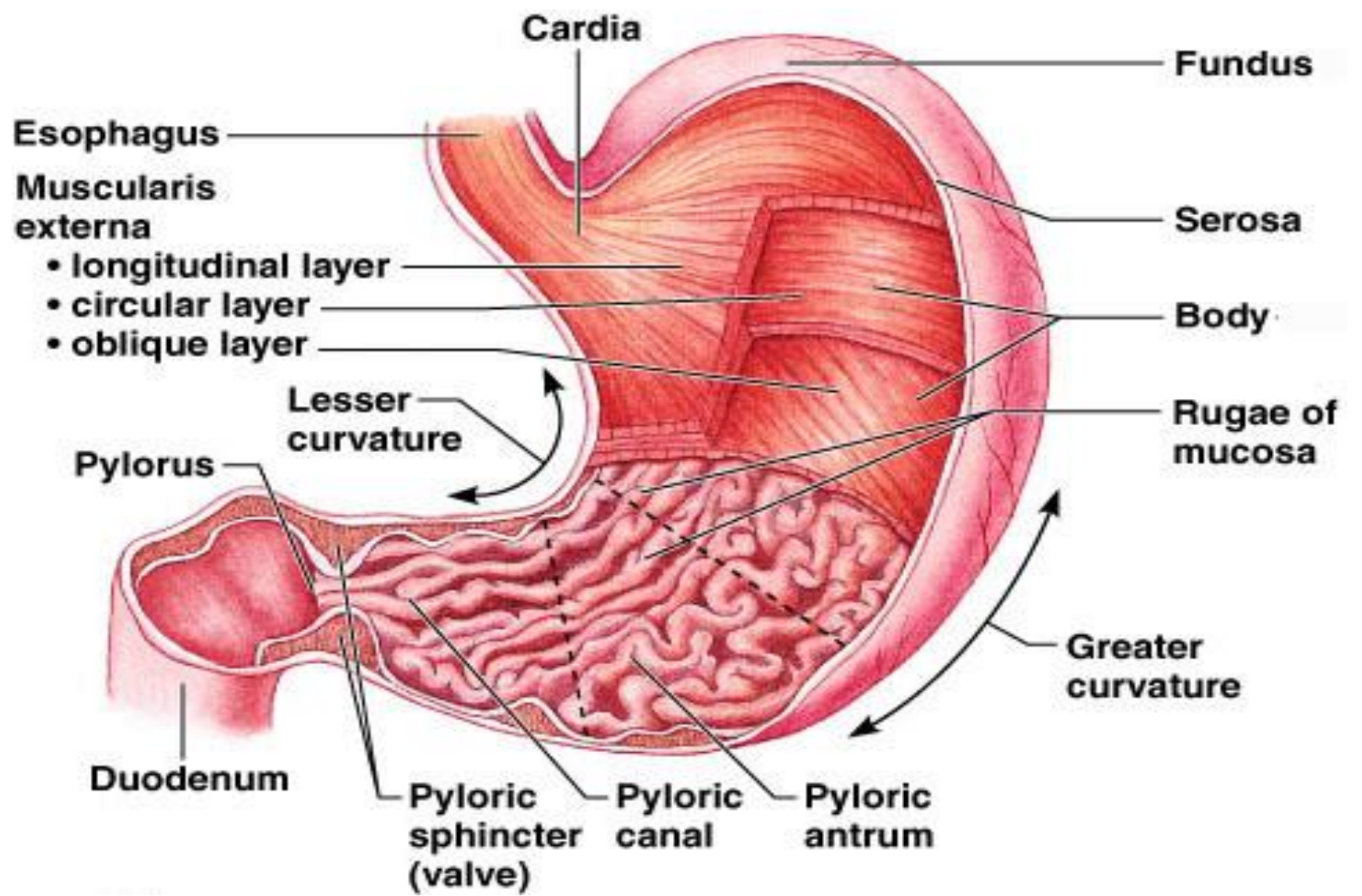


The Stomach

Stomach

- **Functional anatomy:**
 - Cardia → surrounds the cardiac orifice
 - Fundus → dome-shaped region beneath the diaphragm
 - Body → mid-portion of the stomach
 - Pylorus → distal opening of the stomach
 - Pyloric sphincter → continuation of pylorus with the duodenum
 - Role in gastric emptying





Stomach

- Functional anatomy:
 - Muscularis externa
 - Contain longitudinal + circular + oblique muscle layers
 - Allows the stomach to churn, mix, and propel chyme
 - Epithelial lining
 - Contain Goblet cells that produce a coat of **alkaline mucus**
 - The mucous surface layer traps a **HCO₃⁻ rich fluid** beneath it
 - Gastric pits are attached to glands that produce gastric secretions (mucus, HCL, enzymes)

Stomach

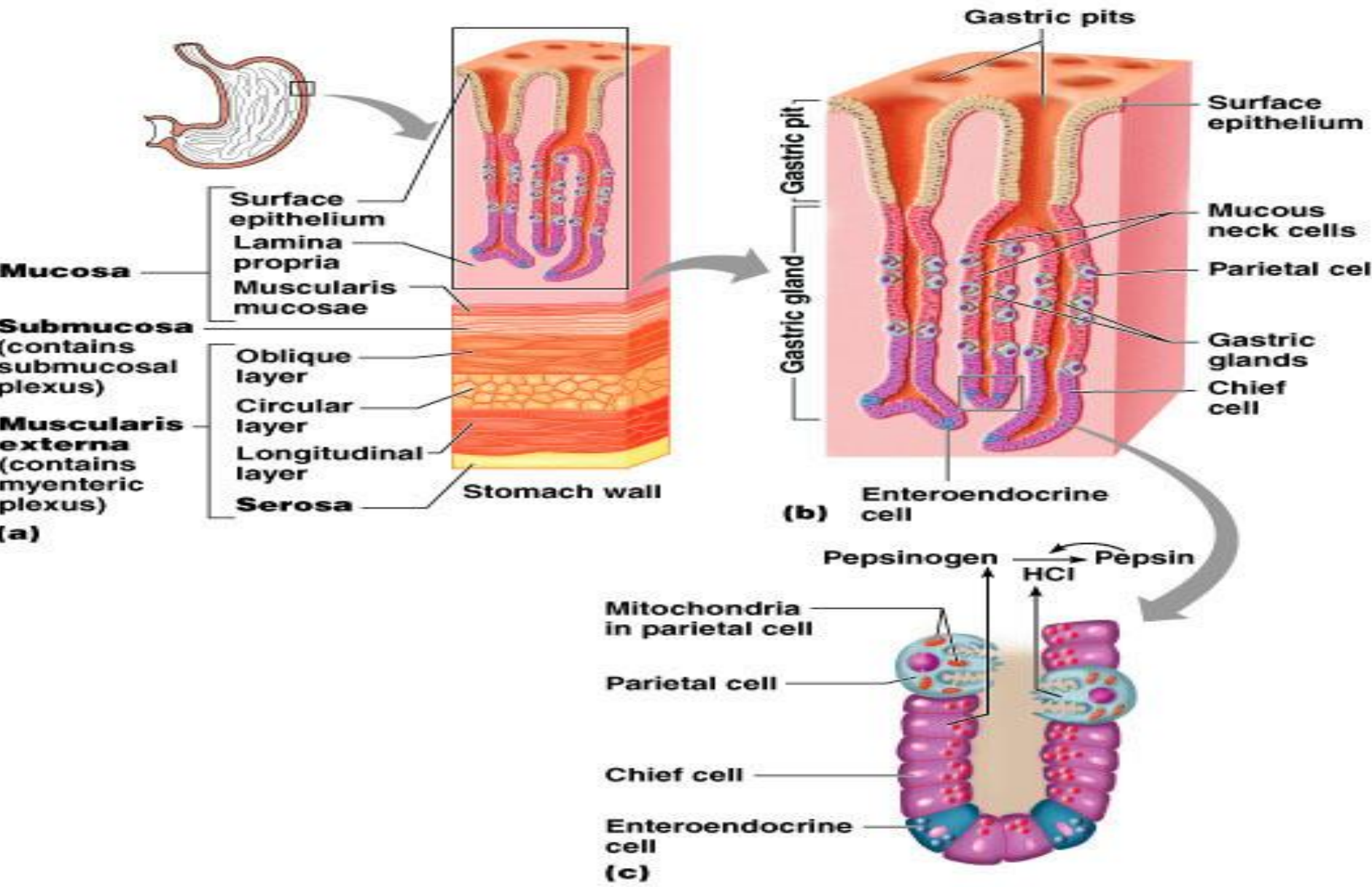
- **Stomach Lining:**
 - Stomach is exposed to the harshest conditions in the digestive tract
 - To prevent HCl from digesting the stomach cells, the stomach has a mucosal barrier with:
 - A thick coat of bicarbonate-rich mucus on the stomach wall
 - Epithelial cells that are joined by tight junctions
 - Gastric glands that have cells impermeable to HCl
 - Damaged epithelial cells are quickly replaced

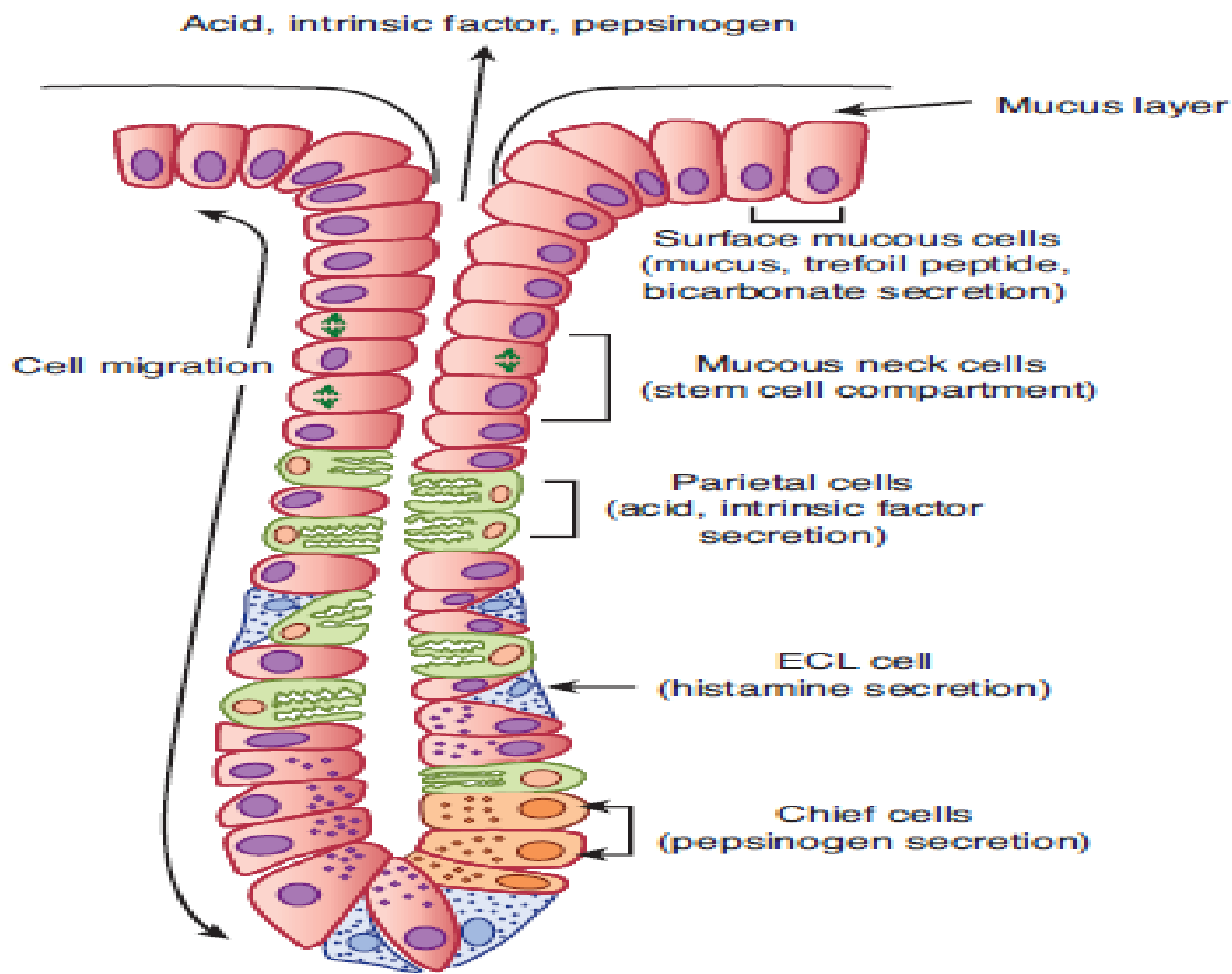
Stomach

- **Functions:**
 - Mixing of bolus with gastric secretions to form semi-fluid chyme
 - Storage of chyme until emptied into the duodenum
 - *Can accommodate up to 1.5L of chyme*
 - Slow emptying of chyme into the small intestine at a rate suitable for proper digestion and absorption by the small intestine
 - Secretory function → HCl, IF, mucous, pepsinogen, gastrin...
 - Sterilization, digestion, absorption
 - Facilitates defecation

Stomach

- **Secretory function:**
 - Mucous neck cells → mucus (*Water + Electrolytes + Glycoproteins + Polysaccharides...*) → Lubrication + Protection
 - Parietal cells → HCl + IF
 - Chief cells → Pepsinogen
 - Pepsinogen is activated to pepsin by HCl in the stomach
 - Pepsin also activates pepsinogen via positive feedback mechanism
 - Enteroendocrine cells → hormone secreting cells in the GI mucosa
 - Enterochromaffin cells → Serotonin + Histamine





Gastric Secretion

- Neural and hormonal factors regulate the release of gastric secretions
- **Phases:**
 - Stimulatory and inhibitory events occur in three phases:
 1. Cephalic (reflex) phase → prior to food entry
 2. Gastric phase → once food enters the stomach
 3. Intestinal phase → as chyme enters the duodenum

Gastric Secretion

1. Cephalic Phase:

- Secretion increased by:
 - Sight or thought of food
 - Stimulation of taste + smell receptors
- Secretion decreased by :
 - Loss of appetite or depression
 - Inhibition of parasympathetic nerves

Gastric Secretion

2. Gastric Phase:

- Secretion increased by:
 - Stomach distension (stretch receptors)
 - Stimulation of chemoreceptors (peptides + caffeine + $\uparrow$ pH...)
 - Release of gastrin to the blood
- Secretion decreased by:
 - Excessive stomach acidity ($\text{pH} < 2$)
 - Emotional upset that overrides the parasympathetic NS

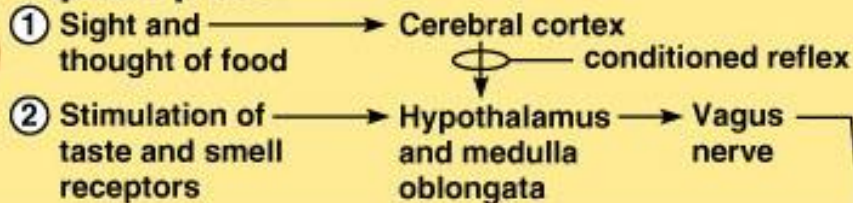
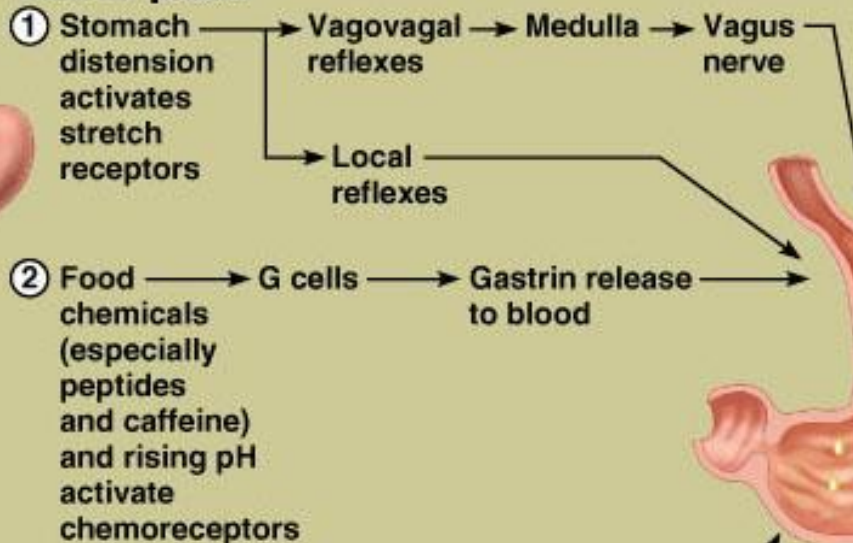
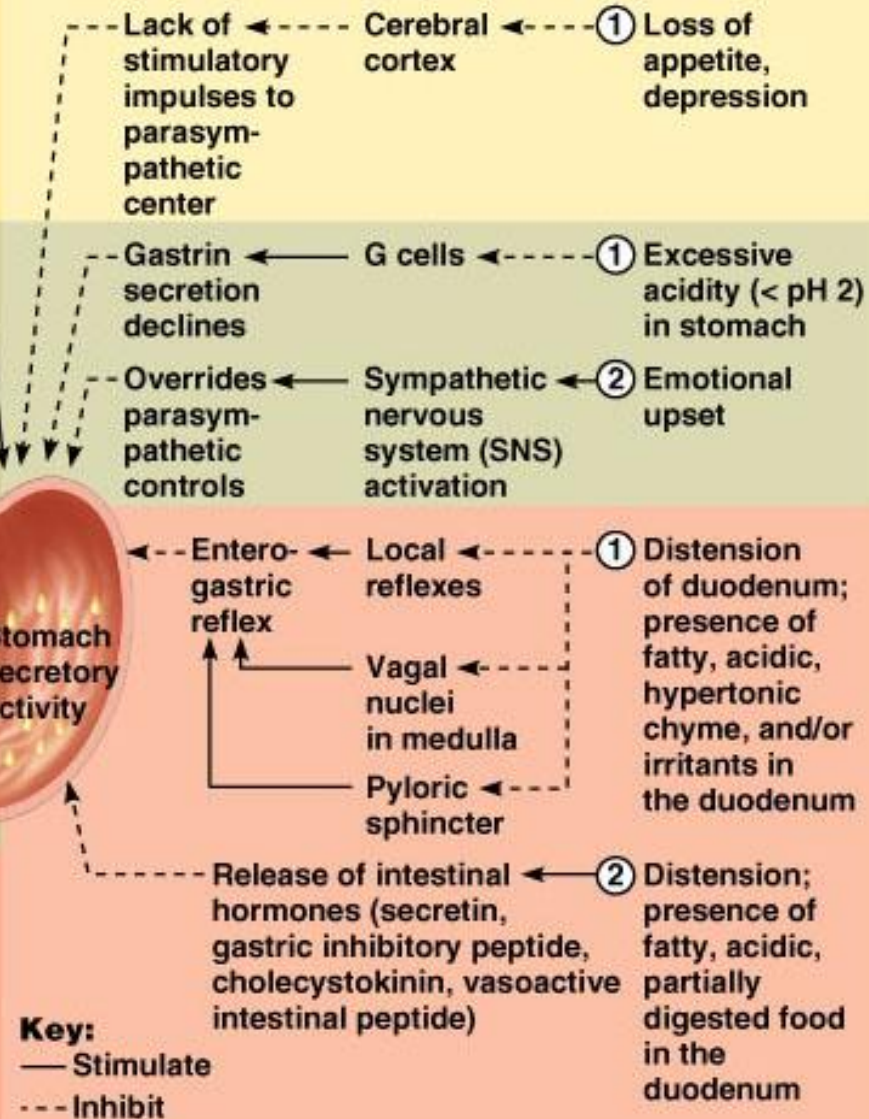
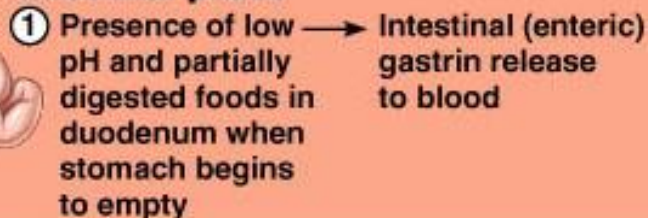
Gastric Secretion

3. Intestinal Phase:

- Secretion increased by:
 - Lower pH
 - Partially digested food in the duodenum which stimulates gastric gland activity
- Secretion decreased by:
 - Distension of the duodenum
 - Presence of fatty, acidic or hypertonic chyme and/or irritants in the duodenum → closes the pyloric sphincter + inhibit gastric secretion

Stimulatory Events

Inhibitory Events

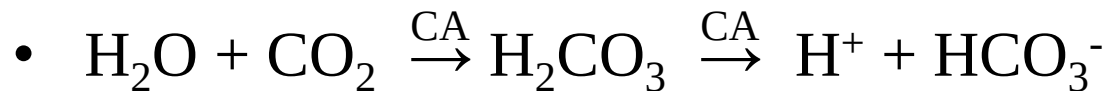
Cephalic phase**Gastric phase****Intestinal phase**

HCL Secretion

- Steps

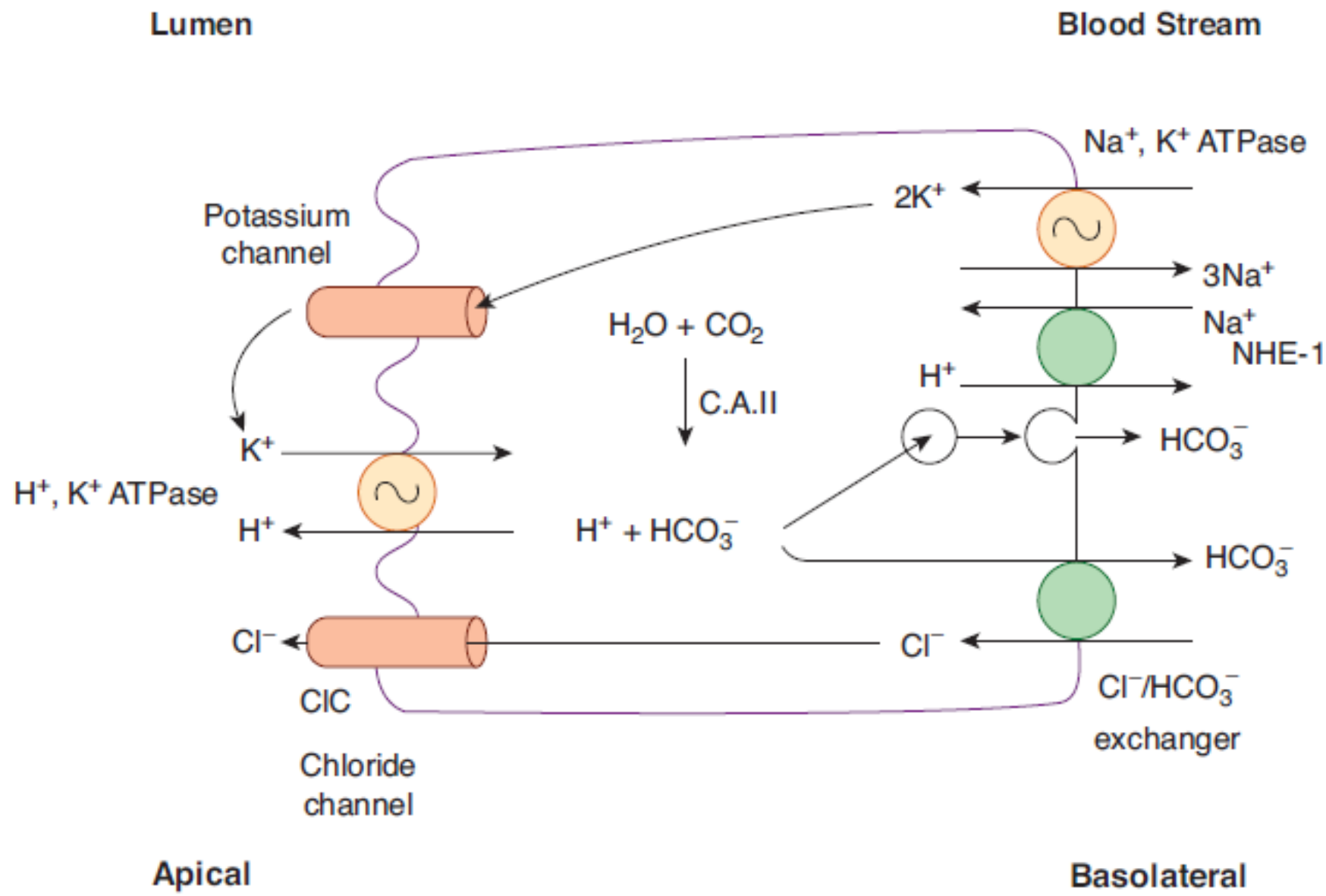
- HCL is secreted by Parietal cells

1. Cl^- is transported into the parietal cells from the ECF in exchange for HCO_3^- and then diffuses into the lumen
2. H^+ is produced from H_2CO_3 and actively pumped into the lumen in exchange for K^+ by H^+-K^+ ATPase

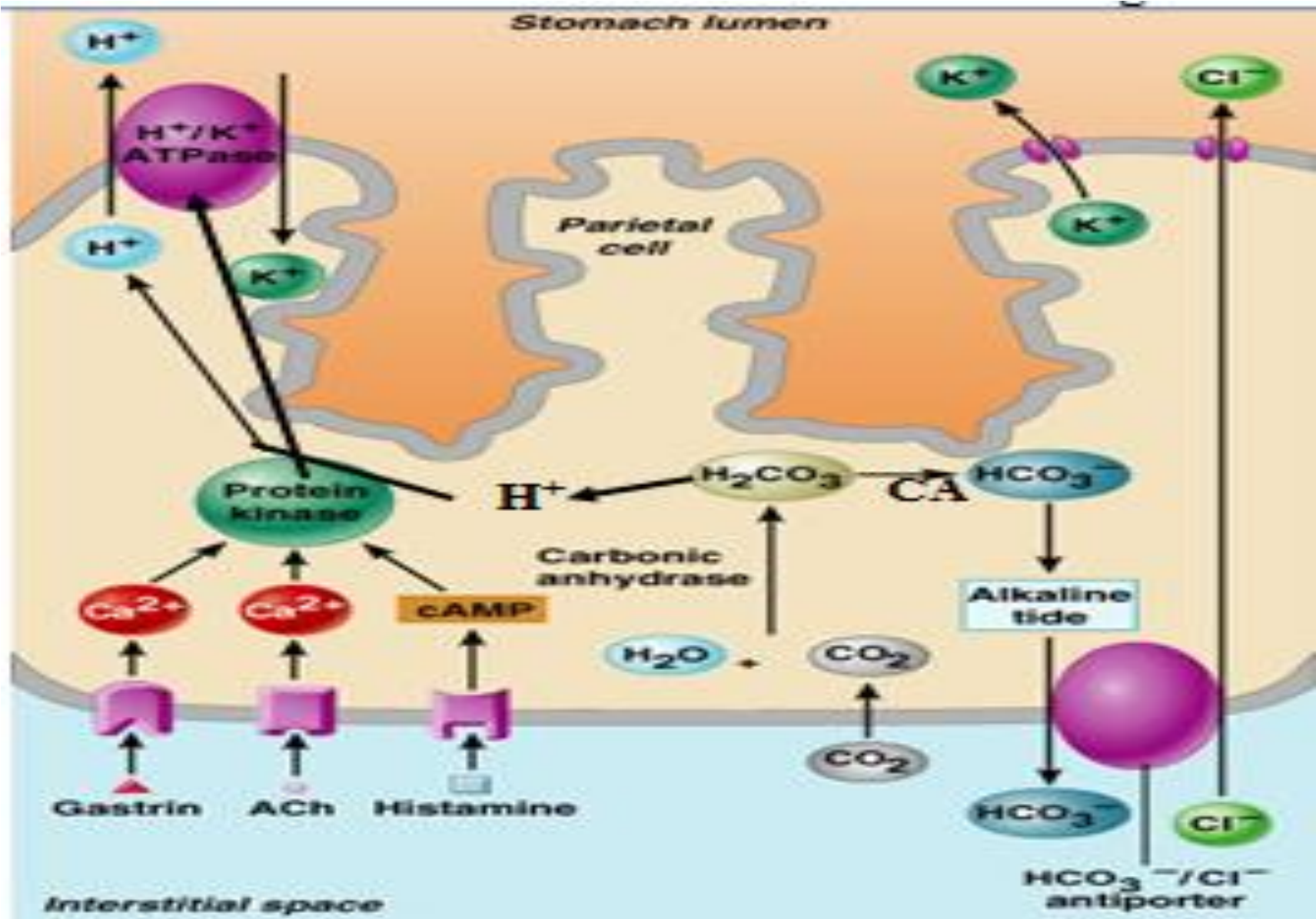


3. Then, Cl^- combines with H^+ in the gastric lumen and forms HCL

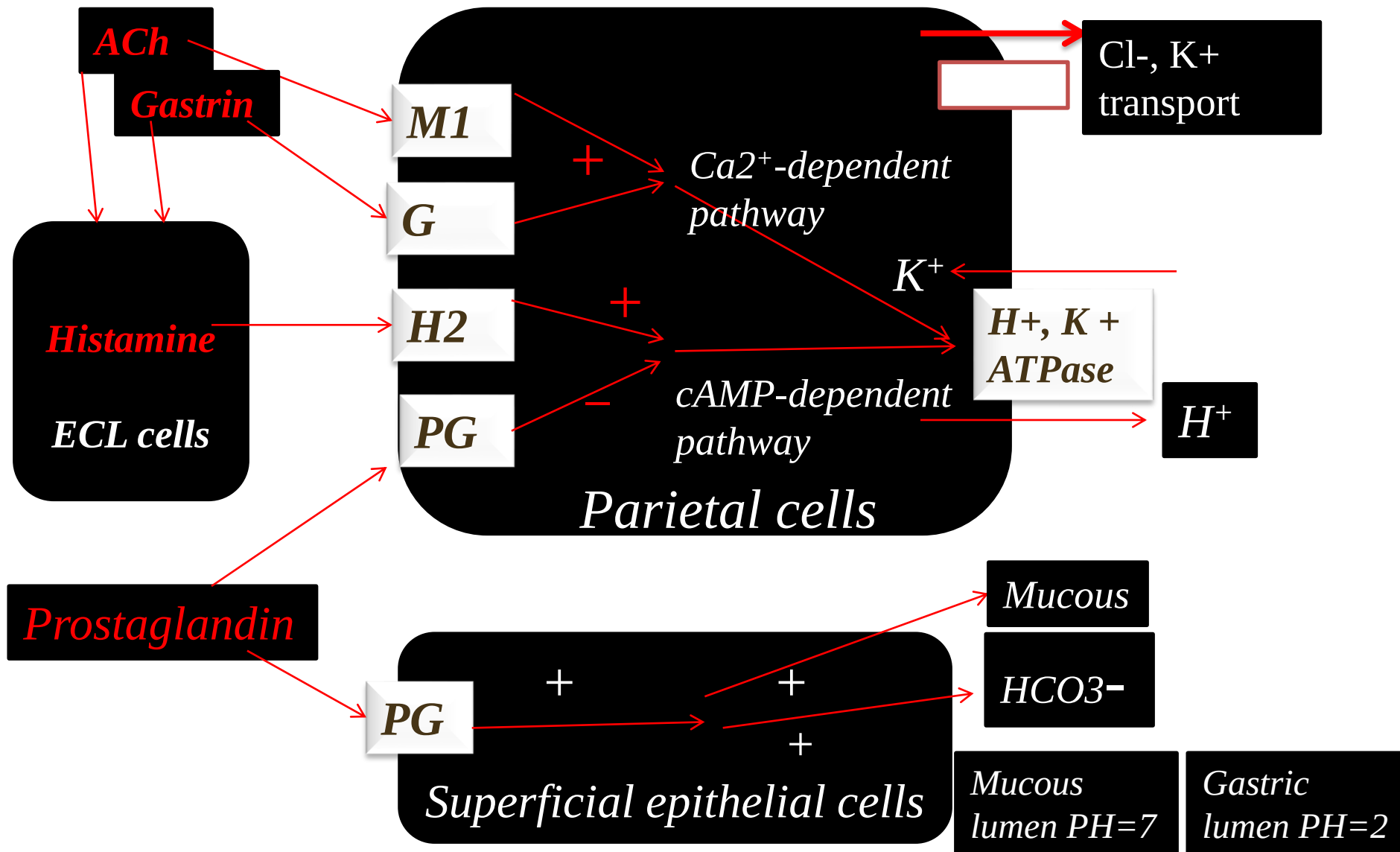
Mechanism of HCL Secretion



Mechanism of HCL Secretion



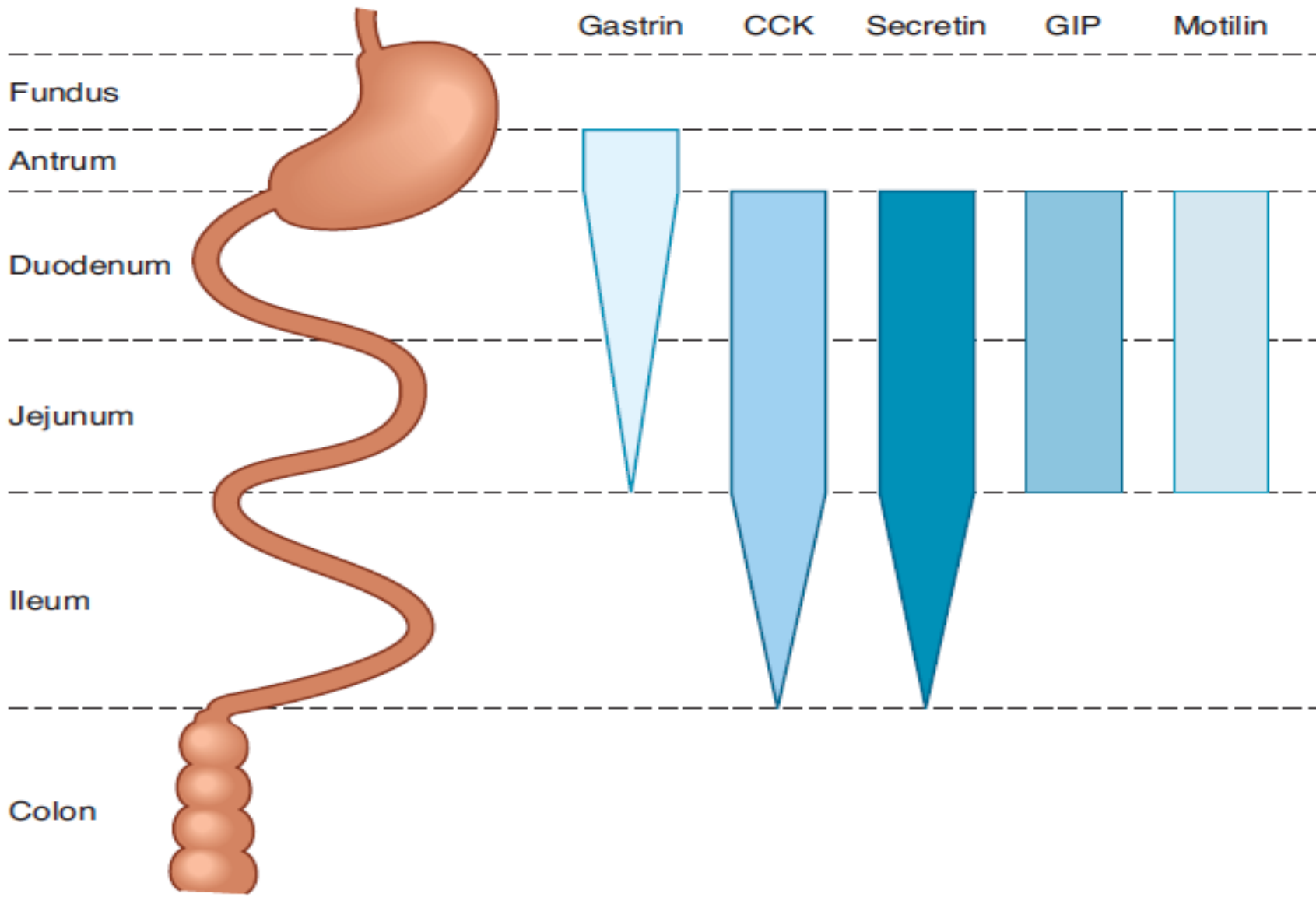
Mechanism of HCl, HCO₃⁻ and mucous production



Gastrin

- Produced by G-cells mainly in the antral portion of the gastric mucosa
- **Functions:**
 - Stimulate HCl and pepsin secretion
 - Stimulate growth of mucosa of the stomach, small and large intestines (trophic action)

Site of production of GI hormones



Gastrin

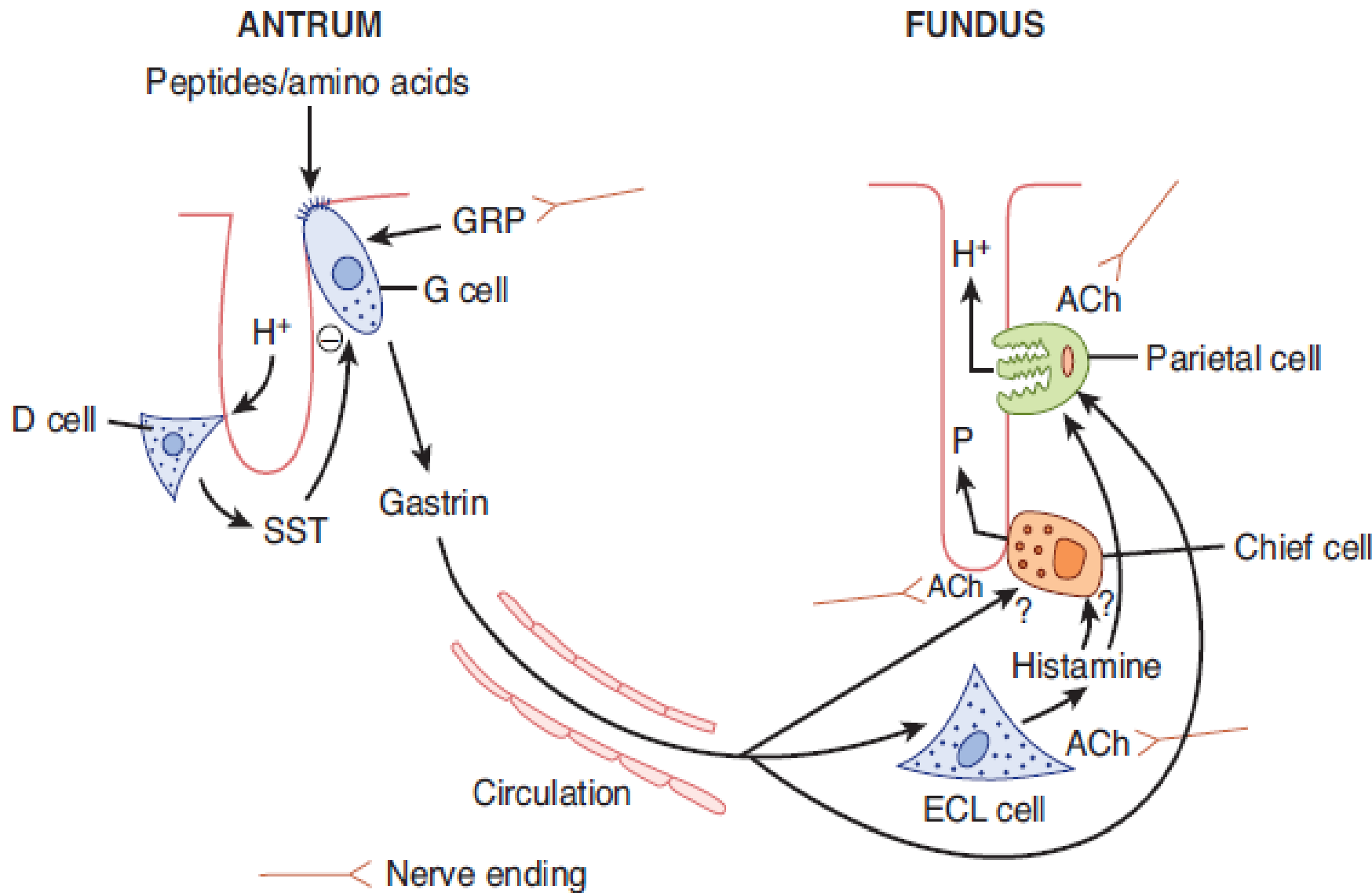
- Factors increasing Gastrin secretion:
 - Peptides and amino acids
 - Stomach distention
 - Increased vagal discharge via GRP (*Gastrin Releasing Polypeptide*)
 - GRP is released by postganglionic vagal fibers that innervate G-cells

Gastrin

- Factors decreasing Gastrin secretion:
 - Gastric acid (*directly and by* $\uparrow$ *somatostatin release*) by –ve feedback
 - Somatostatin $\rightarrow$ potent inhibitor of gastrin secretion
 - Secretin, GIP, VIP, glucagon, calcitonin

NB:

- Gastrin secretion is chronically elevated if parietal cells are damaged



Cholecystokinin (CCK)

- Secreted by I-cells in the upper small intestine
- CCK is found in nerves in the distal ileum and colon
- CCK receptors:
 - CCK-A (periphery & brain) + CCK-B (brain)
 - Regulation of food intake + anxiety + analgesia
- Presence of FAs + peptides & AAs in the intestine + CCK-releasing peptide (intestinal mucosa) increases CCK secretion
 - Digestion products → CCK secretion → bile + pancreatic secretion → further digestion → CCK secretion (+ve feedback)

Cholecystokinin

- **Functions:**
 - Stimulate pancreatic enzyme secretion
 - Stimulate contraction of the gallbladder
 - Relaxation of the sphincter of Oddi, which allows both bile and pancreatic juice to flow into the intestinal lumen
 - Inhibit gastric emptying
 - Enhance motility of the small intestine and colon
 - Exert trophic effect on the pancreas

Cholecystokinin

- Functions:
 - CCK + Secretin:
 - Augment contraction of the pyloric sphincter → prevent the reflux of duodenal contents into the stomach.
 - Augment production of alkaline pancreatic juice
 - CCK + Gastrin:
 - Stimulate glucagon secretion

Secretin

- Secreted by S-cells in the deep mucosa of upper small intestine
- Secretion is increased by digestion products of proteins + emptying of gastric acid to the upper intestine
- **Functions:**
 - Increase bicarbonate secretion by the pancreas and biliary tract
 - Augment the action of CCK in producing pancreatic enzymes
 - Decreases gastric acid secretion
 - Cause contraction of the pyloric sphincter

Gastric Inhibitory Peptide (GIP)

- Produced by K-cells in the mucosa of the duodenum and jejunum
- Secretion is stimulated by glucose and fat in the duodenum
- Inhibits gastric secretion and motility at large doses
- Stimulates insulin secretion (*glucose-dependent insulinitropic polypeptide*)

Vasoactive Intestinal Peptide (VIP)

- Found in the GIT nerves + brain + autonomic nerves + in the blood
- Markedly stimulates intestinal secretion of electrolytes and water
- Relax intestinal smooth muscle (+ sphincters)
- Dilate peripheral blood vessels
- Inhibit gastric acid secretion
- Potentiate the action of acetylcholine in the salivary glands

Motilin

- Secreted by ECF cells and Mo-cells in the stomach & SI
- Acts on G-protein coupled receptors on enteric neurons
- Cause contraction of smooth muscles in the stomach and intestines
- A major regulator of the migrating motor complexes (MMCs) that control GI motility between meals
- When a meal is ingested, secretion of motilin is suppressed until digestion and absorption are complete
- Its circulating level increases at intervals in the interdigestive state

Somatostatin

- Secreted by D-cells of the GI mucosa + pancreatic islets + hypothalamus
- Secretion is stimulated by increased acidity in the lumen:
 - *Inhibit gastrin secretion produced by the acid*
- It also inhibits:
 - Secretion of VIP, GIP, secretin, motilin, pancreatic secretion; gastric acid secretion and motility and gallbladder contraction
 - Absorption of glucose, amino acids, and triglycerides

Other GI Secretions

Peptide YY:

- Secreted from the jejunum
- Inhibits gastric acid secretion and motility
- Its release is stimulated by fat

Ghrelin:

- Secreted primarily by the stomach
- Play an important role in the central control of food intake (+)
- It also stimulates growth hormone secretion by acting directly on receptors in the pituitary

Other GI Secretions

Substance P:

- Found in endocrine and nerve cells in the GI tract
- It increases motility of the small intestine

Guanylin:

- A peptide secreted by GI mucosal cells (*from pylorus to rectum*)
- Stimulate GC $\rightarrow \uparrow$ cGMP $\rightarrow \uparrow$ secretion of Cl⁻ to intestinal lumen
- **NB** $\rightarrow$ certain strains of E. coli has a very similar structure to guanylin

Gastric Emptying

- Gastric emptying is regulated by:
 - Neural → Entero-gastric reflex
 - Hormonal mechanisms
- CHO-rich chyme quickly moves through the duodenum
- Fat-rich chyme is digested more slowly causing food to remain in the stomach longer

Gastric Emptying

Stimulatory stomach factors

Distention of the stomach

Partially digested protein

Alcohol

Caffeine

↑Gastrin
Secretion

Sensory
impulse via vagus

Constrict LES

↑Stomach motility

Relax pyloric sphincter

Stimulate gastric emptying

Inhibitory duodenal factors

Distention of the duodenum

Fatty acids and glucose

Partially digested protein

↑Secretion of

CCK

GIP

Secretin

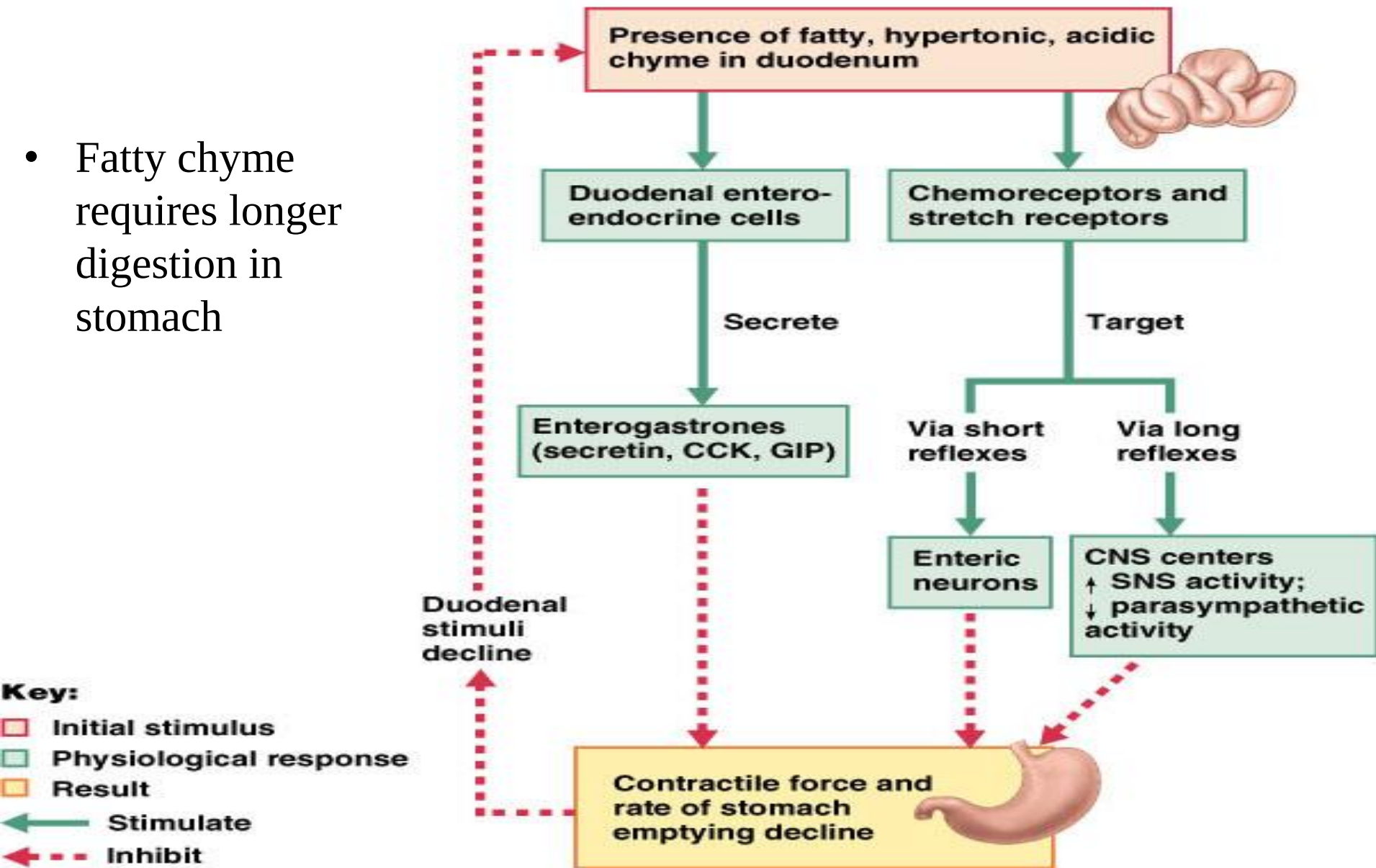
Entero-gastric reflex

↓Stomach motility

↑Pyloric sphincter tone

Inhibit gastric emptying

- Fatty chyme requires longer digestion in stomach



The Small Intestine

Small Intestine

- Extends between the pyloric sphincter and the Ileocecal valve
- The longest part of the GI tract $\approx$ 3m long ($\approx$ 2x in a cadaver)
- Has 3 basic portions:
 - Duodenum $\approx$ the 1st 20-30cm portion of the SI
 - Jejunum $\approx$ the next 2/5 portion of the SI
 - Ileum $\approx$ the last 3/5 portion of the SI
 - The ileum empties into the large intestine through the ileocecal valve

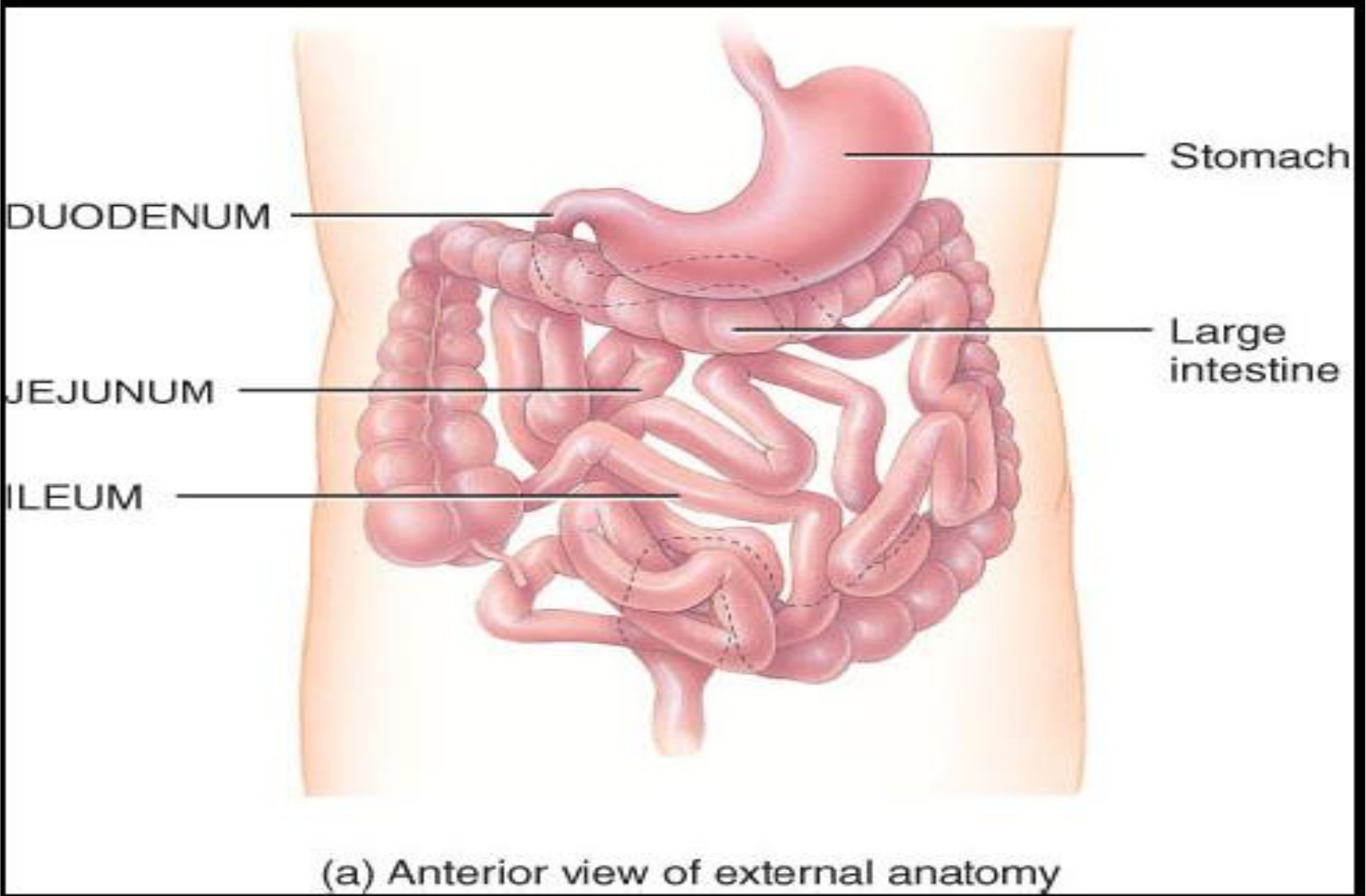
Small Intestine

1. Duodenum:

- A hollow jointed tube connecting the stomach to the jejunum
- The first and shortest part of the small intestine
- The entry point of bile and pancreatic juices

2. Jejunum:

- Located b/n distal end of duodenum and proximal part of ileum
- Its mucous membrane is covered with villi
- Have fewer goblet cells and generally lacks Peyer's patches



Small Intestine

3. Ileum:

- Lined with villi + microvilli (*Large surface area for majority of absorption*)
- The villi contain large numbers of capillaries that take the amino acids and glucose produced by digestion to the hepatic portal vein and to the liver
- Contain **protease** and **carbohydrate enzymes** responsible for the final stages of protein and CHO digestion
- Absorb **vitamin B12** and **bile salts**

Small Intestine

- Villi/Microvilli
 - Mucosa of the SI is folded into villi that project into the lumen
 - They are covered with columnar epithelial cells (*interspersed mucus-secreting goblet cells*)
 - The lamina propria forms the connective tissue core of each villus:
 - Contains lymphocytes + capillaries + lymphatic vessels (central lacteal)
 - Monosaccharides and amino acids enter the blood capillaries
 - Fats enter the central lacteals

Intestinal Secretion

- Brunner's gland:
 - Located in the submucosa of the duodenum.
 - Secrete alkaline fluid containing mucin, which protects the mucosa from the acidic stomach contents entering the duodenum
- Goblet cells → secrete mucin and create a protective mucus layer
- Crypts of Lieberkühn:
 - Tubular glands formed from the mucosa of the small intestine in between the bases of the villi.
 - Secrete intestinal juice (mucous and electrolytes)

Intestinal Secretion

- Enteroendocrine cells → produce hormones
- Enterochromaffin cells → serotonin producing cells
- Enterocytes → secrete digestive enzymes

Intestinal Secretion

- Digestive enzymes:
 - Plasma membranes of the microvilli contain digestive enzymes (hydrolyze disaccharides, polypeptides...) → facilitate digestion
 - These enzymes remain attached to the cell membrane with their active sites exposed to the chyme (*didn't secrete into the lumen*)
 1. Peptidase → splits peptides into amino acids
 2. Sucrase + Maltase + Lactase → hydrolyse disaccharides into monosaccharides
 3. Intestinal lipase → splits neutral fats into glycerol and fatty acids

Intestinal Secretion

1. Peptidases:

- **Enterokinase** activates **trypsin** (pancreas) + indirectly other pancreatic enzymes which digest proteins
- **Aminopeptidase** → Produces free aas, dipeptides, and tripeptides
 - Deficiency results in **protein malnutrition**

Intestinal Secretion

2. Dissacharidases:

- Sucrase $\leftrightarrow$ sucrose $\rightarrow$ glucose and fructose
- Maltase $\leftrightarrow$ maltose $\rightarrow$ glucose
- Lactase $\leftrightarrow$ lactose $\rightarrow$ glucose + galactose
 - Deficiency causes **Lactose intolerance**

3. Phosphatases:

- $\text{Ca}^{2+} \text{ Mg}^{2+}$ -ATPase $\rightarrow$ absorption of Ca^{2+} (Regulated by vitamin D)
- Alkaline phosphatase $\rightarrow$ remove phosphate groups from organic molecules

Intestinal Secretion

- **NB**
 - **Digestion of lactose** (milk sugar) depends on the presence of lactase
 - Lactase is present in all children (< 4years of age) (becomes inactive to some degree in most adults)
 - *(Asian & African's are more often lactase deficient than Caucasians)*
 - Deficiency of lactase can result in **lactose intolerance** → too much undigested lactose in the intestine → Causes diarrhea, gas, cramps...
 - **Yogurt** is better tolerated than milk *(it contains lactase produced by the yogurt bacteria)*

Intestinal Secretion

- Regulation of SI secretion:
 - Local factors → tactile, distension, irritation, pH
 - Hormonal → Secretin + CCK + VIP + Glucagon + GIP
 - Neural:
 - Vagal stimulation → increase intestinal secretion
 - Sympathetic stimulation → decrease intestinal secretion

Small Intestine

- **Absorption:**
 - Nutrients are absorbed across the epithelium of intestinal mucosa
 - CHO's, lipids, aa's, Ca^{2+} & Fe $\rightarrow$ primarily absorbed in the duodenum and jejunum
 - Bile salts, vit B12 $\rightarrow$ absorbed primarily in the ileum
 - Absorption is rapid due to extensive folding of the intestinal mucosa (*Villi/microvilli $\rightarrow$ greatly increase its absorptive surface area*)
 - Mucosa and submucosa form large folds (plicae circulares)

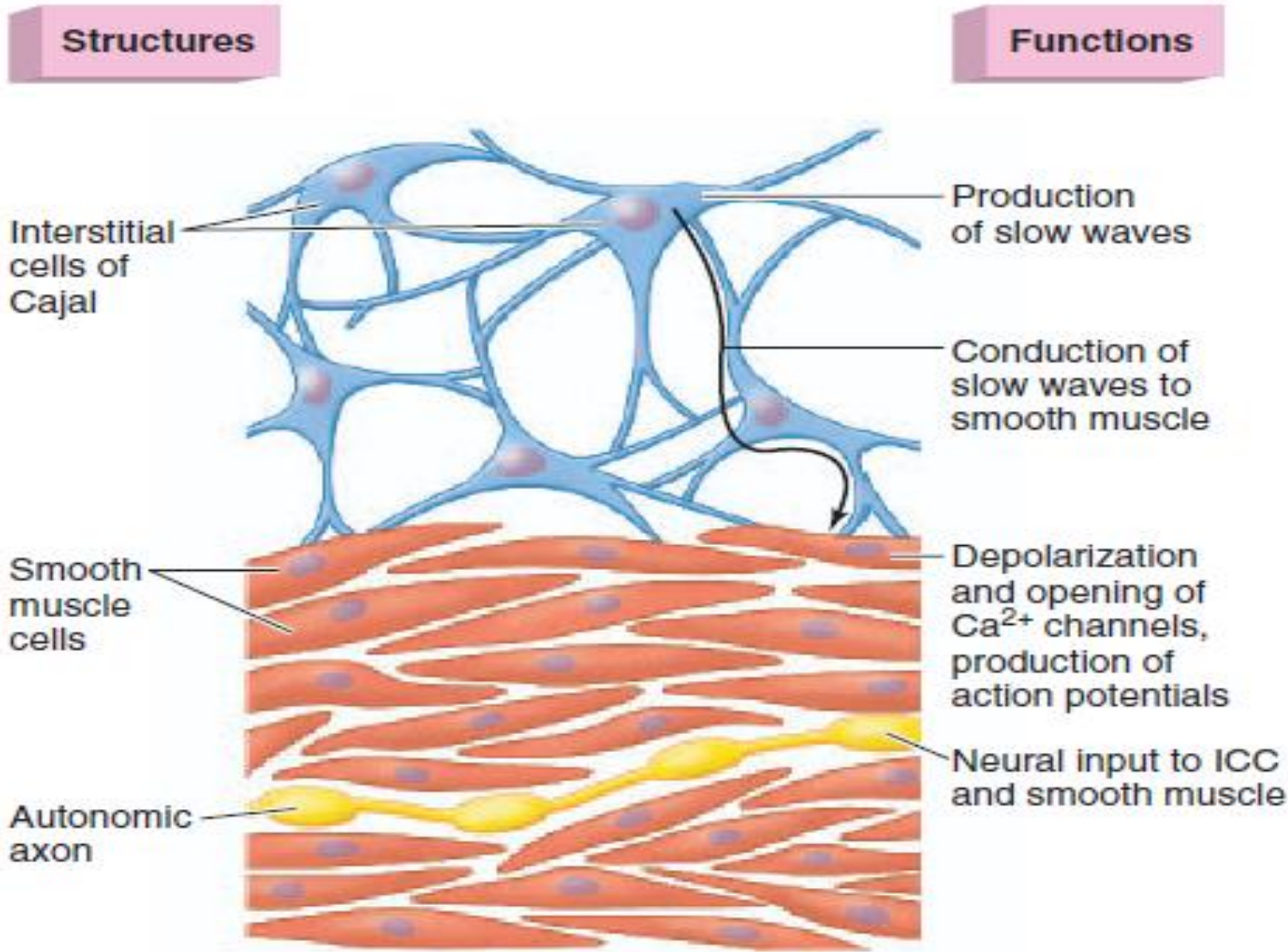
Small Intestine

- **Absorption:**
 - Most of the fluid and electrolytes are absorbed by the small intestine
 - Absorption of water occurs passively as a result of the osmotic gradient created by the active transport of ions
 - Mucosal epithelial cells contain Na^+/K^+ pumps in the basolateral membrane
- **NB**
 - Aldosterone stimulate salt and water absorption in the ileum

Intestinal Motility

- **Interstitial cells of Cajal** are joined to each other and to smooth muscle cells by gap junctions:
 - Conduct slow waves between interconnected interstitial cells of Cajal
- The slow waves serve to depolarize adjacent smooth muscle cells
- When the slow-wave depolarization exceeds threshold, it triggers APs in the smooth muscle cells by opening VG Ca^{2+} channels

Intestinal Slow waves



Intestinal Motility

- Influx of Ca^{2+} has two effects:
 1. Depolarize intestinal smooth muscles
 - Repolarization is caused by the efflux of K^{+}
 2. Causes contraction
 - Contraction is aided by additional Ca^{2+} released from the SR through Ca^{2+} induced Ca^{2+} release

Intestinal Motility

- Autonomic nerves modify these automatic contractions
 - Ach (from PNS axons) depolarizes and increases the duration of slow waves → promotes motility of the intestine
 - NE hyperpolarize the smooth muscle membrane → decrease the activity of the intestine
- Movement of chyme through the intestine is relatively slow
- Caused primarily by the greater pressure at the pyloric end of the SI than at the distal end

Intestinal Motility

- Two major types of contractions occur in the small intestine:
 1. Segmentation (mixing movement)
 - The major contractile activity of the SI
 2. Propulsive movement
 - Much weaker in the SI than in the esophagus and stomach

Intestinal Motility

1. Segmentation:

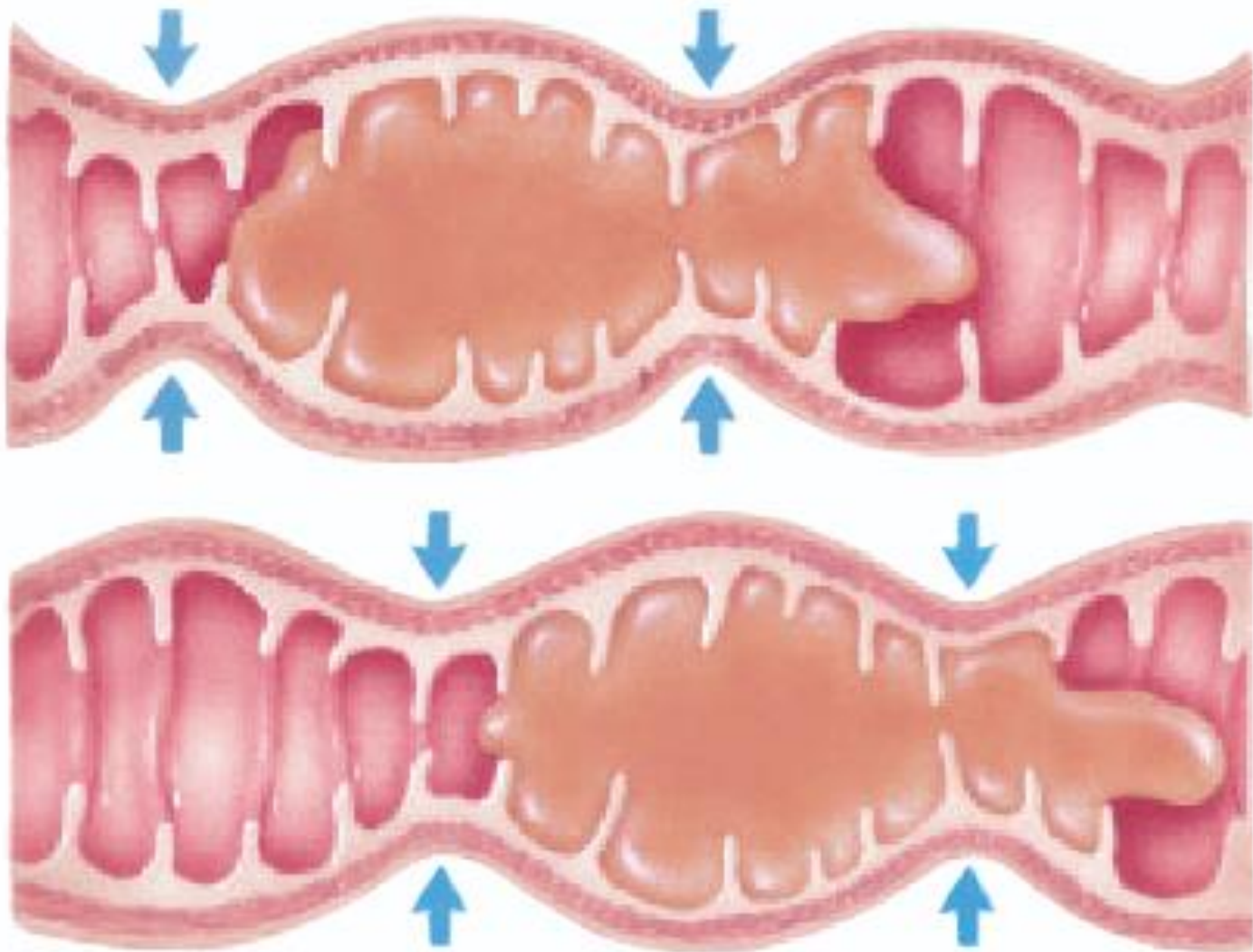
- Ring like contractions which appear at regular intervals with 2-3cm distance
- Repeated at a rate of 12cycles/min in duodenum and 8/min in ileum
- Exert chopping action on intestinal chyme and mix it with digestive juice

Intestinal Motility

1. Segmentation:

- Muscular constrictions of the lumen, which occur simultaneously at different intestinal segments → serves to mix the chyme more thoroughly
- Contractions of intestinal smooth muscles occur automatically in response to endogenous pacemaker activity
- The rhythm of contractions is paced by graded depolarizations (*slow waves*) → produced by *interstitial cells of Cajal* (*associated with autonomic nerve endings*)

Intestinal Segmentation



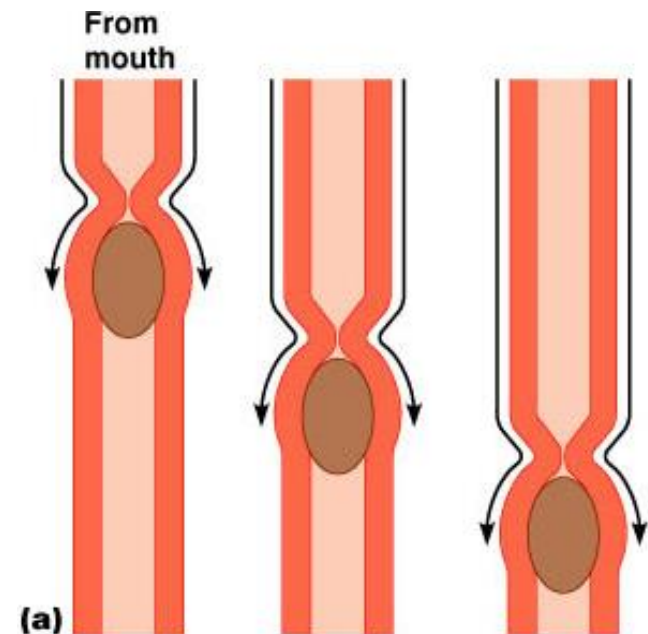
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Intestinal Motility

2. Propulsive movement

- Peristaltic waves that propel chyme anal wards
- Initiated by intestinal distension
- Unabsorbed substances will be propelled to the LI through the ileocecal sphincter

- **NB**
 - Peristaltic rush → diarrhoea



Intestinal Motility

- Ileocecal sphincter
 - Prevent back flow of fecal matter from the cecum to the ileum
- Factors regulating the sphincter:
 - Pressure and chemical irritation of ileum relax the sphincter and empty ileal contents to the cecum
 - Pressure and chemical irritation of cecum inhibit peristalsis of ileum and closes the sphincter
- **NB**
 - **Paralytic ileus** → decreased peristaltic activity in the SI → SI contents are not propelled into the colon

Intestinal Motility

- Regulation of SI Motility:
 - Intestinal motility depends on distension which sends impulses to the enteric plexus & CNS
 - Presence of chyme initiate enteric reflexes → increase intestinal motility
 - Distension produces more vigorous peristalsis $\approx 10\text{cm/s}$
- **NB**
 - PNS signals increase motility, SNS signals decrease motility

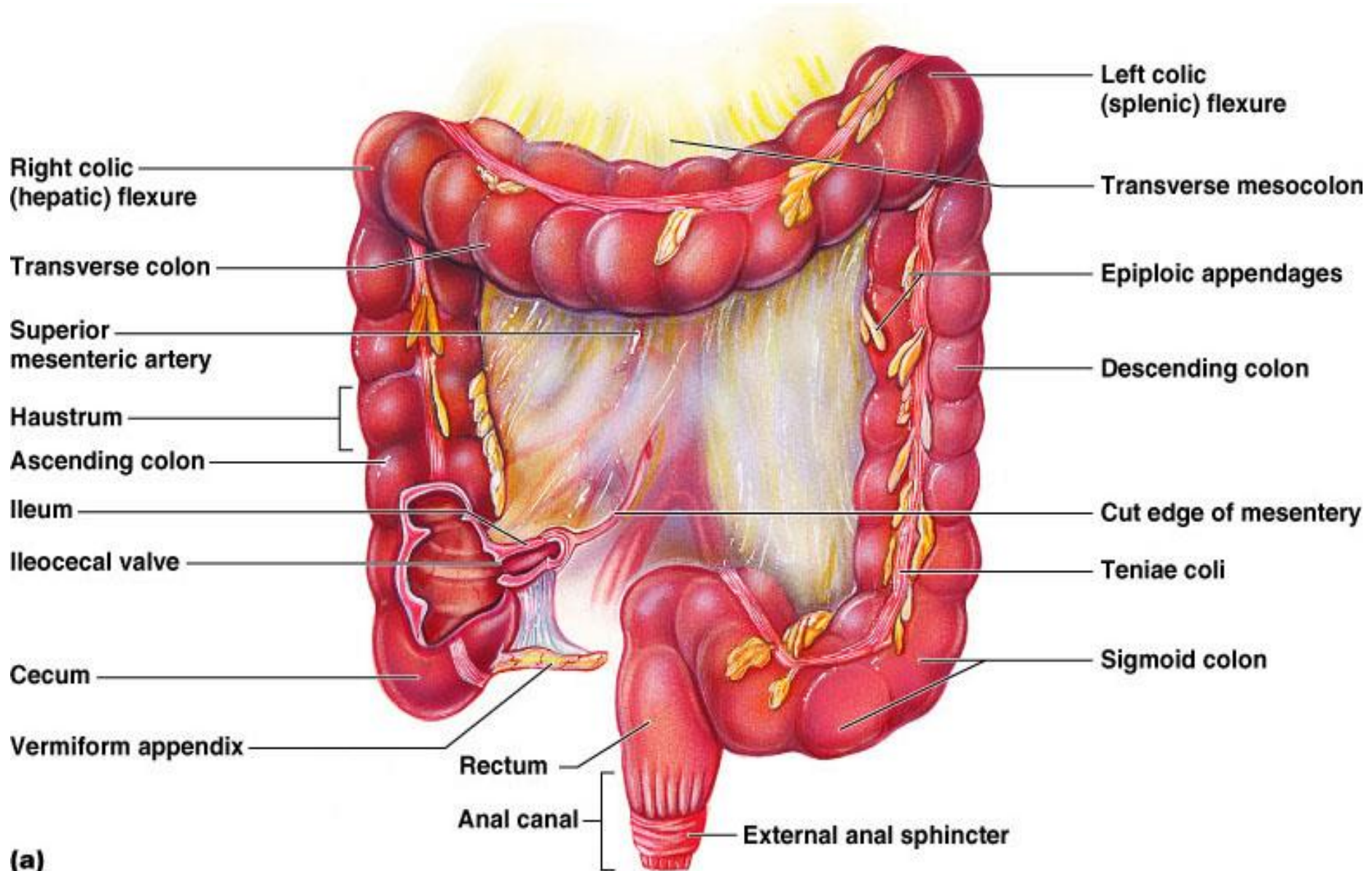
The Large Intestine

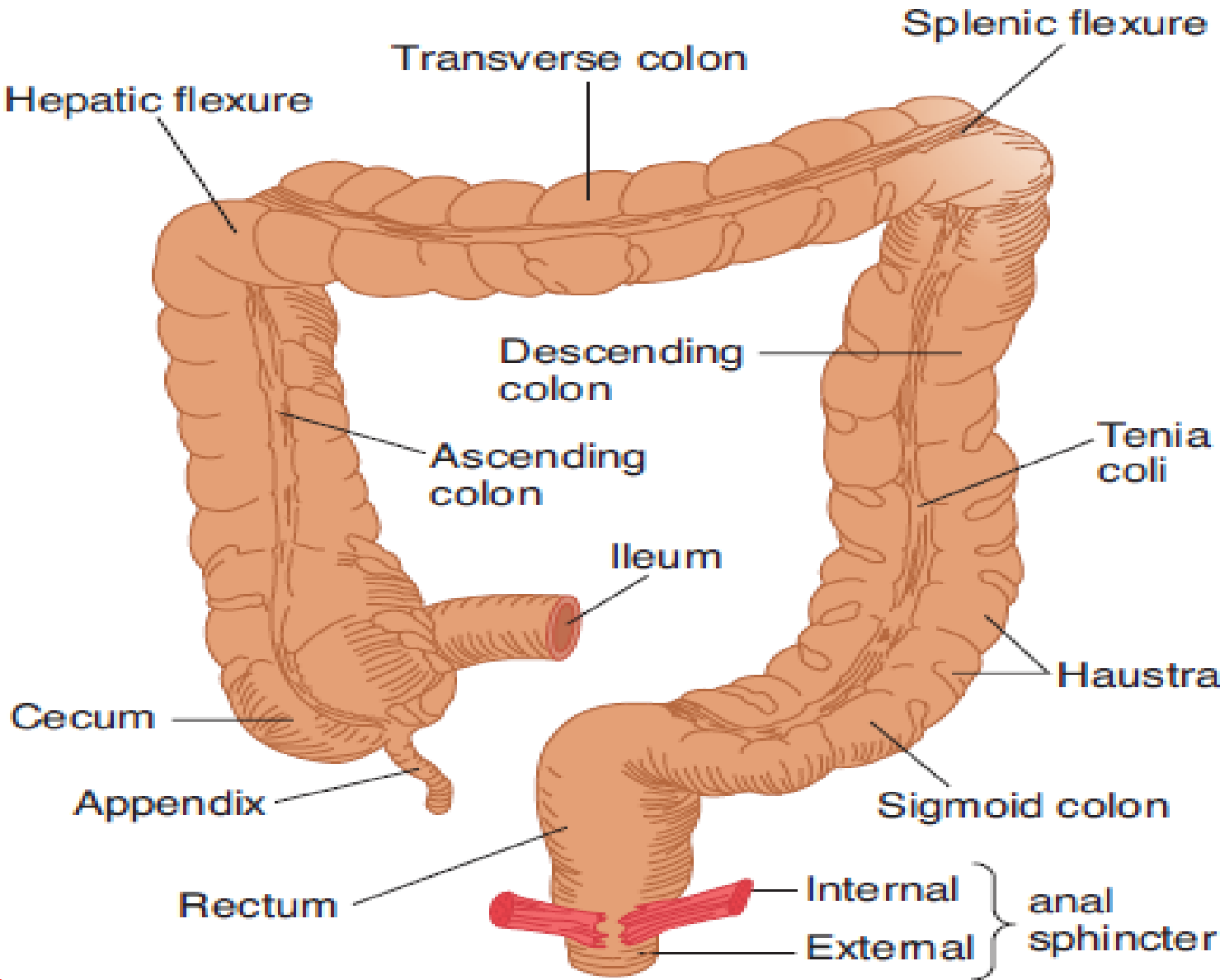
Large Intestine

- Extends from the ileocecal valve to the anus
 - Divided into the cecum + colon + rectum + anal canal
- The mucosa is covered by columnar epithelial cells and mucus secreting goblet cells (but no villi)
- It also contains lymphocytes and lymphatic nodules
- Outer surface of the colon bulges outward to form pouches (haustra)
- The muscularis externa of the haustra occasionally become weakened and the wall forms a more elongated outpouching (diverticulum)
 - *Diverticulitis* → *Inflammation of these structures*

Large Intestine

- The large intestine has little or no digestive function
- Waste material then passes in sequence through:
 - Ascending colon → transverse colon → descending colon → sigmoid colon → rectum → anal canal → anus → excretion
- Absorbs water and electrolytes + vit-B complex and vit-K
 - The SI passes 1.5-2L/day of fluid to the LI and $\approx 90\%$ of this fluid is absorbed in the LI
- Secretion of water occurs by osmosis as a result of active absorption of Na^+/Cl^- out of the epithelial cells into the intestinal lumen





Colon

Large Intestine

- **Intestinal microflora** (colonic bacteria):
 - Produce vit-B, vit-K and folic acid
 - Ferment (anaerobic respiration) some indigestible molecules in the chyme and secreted mucus
 - Produce short-chain fatty acids:
 - Used for energy by the epithelial cells of the colon
 - Aid absorption of Na^+ , Ca^{2+} , Mg^{2+} , HCO_3^- , Fe in the LI
 - Breakdown toxins and stop proliferation of harmful organisms...

Large Intestine

- Types of movements:
 1. Mixing movement (Haustration)
 2. Propulsive movements (mass movement) → initiated by local distension, gastro-colic reflex
- **Poor motility** of the transverse colon causes greater absorption and constipation
- **Excess motility** of the sigmoid colon causes less absorption and diarrhoea or loose stool

Large Intestine

- Secretions:

- Glands
 - Crypts of Lieberkuhn
 - Goblet cells
- } Secret H_2O + electrolytes + mucus
- Regulated by local (tactile) factors

Defecation

- Distention of the stomach by food initiates contractions of the rectum and a desire to defecate (*Gastrocolic reflex*)
 - Amplified by the action of gastrin on the colon
- The urge to defecate first occurs when rectal pressure increases to about 18mmHg
- When this pressure reaches 55mmHg, the external and internal anal sphincters relax → reflex expulsion of contents of the rectum
 - *This is why reflex evacuation of the rectum can occur even in the setting of spinal injury*

Defecation

- **Internal anal sphincter:**
 - Sympathetic (excitatory) and
 - Parasympathetic (inhibitory)
 - Relaxes when the rectum is distended
- **External anal sphincter:**
 - **Pudendal nerve** (skeletal muscle)
 - Maintained in a state of tonic contraction
 - Moderate distention of the rectum increases the force of its contraction
 - Before the pressure that relaxes the external anal sphincter is reached, voluntary defecation can be initiated by straining

Defecation

- Rectal distension → stimulation of myenteric plexus → initiates peristaltic waves in descending colon, sigmoid colon and rectum → peristaltic wave forces feces to the anus → Defecation
 - Inhibition of IAS by PNS + myenteric plexus
 - Voluntary relaxation of EAS
- **NB:**
 - Defecation is a spinal reflex that can be voluntarily inhibited by keeping the external sphincter contracted or facilitated by relaxing the sphincter and contracting the abdominal muscles

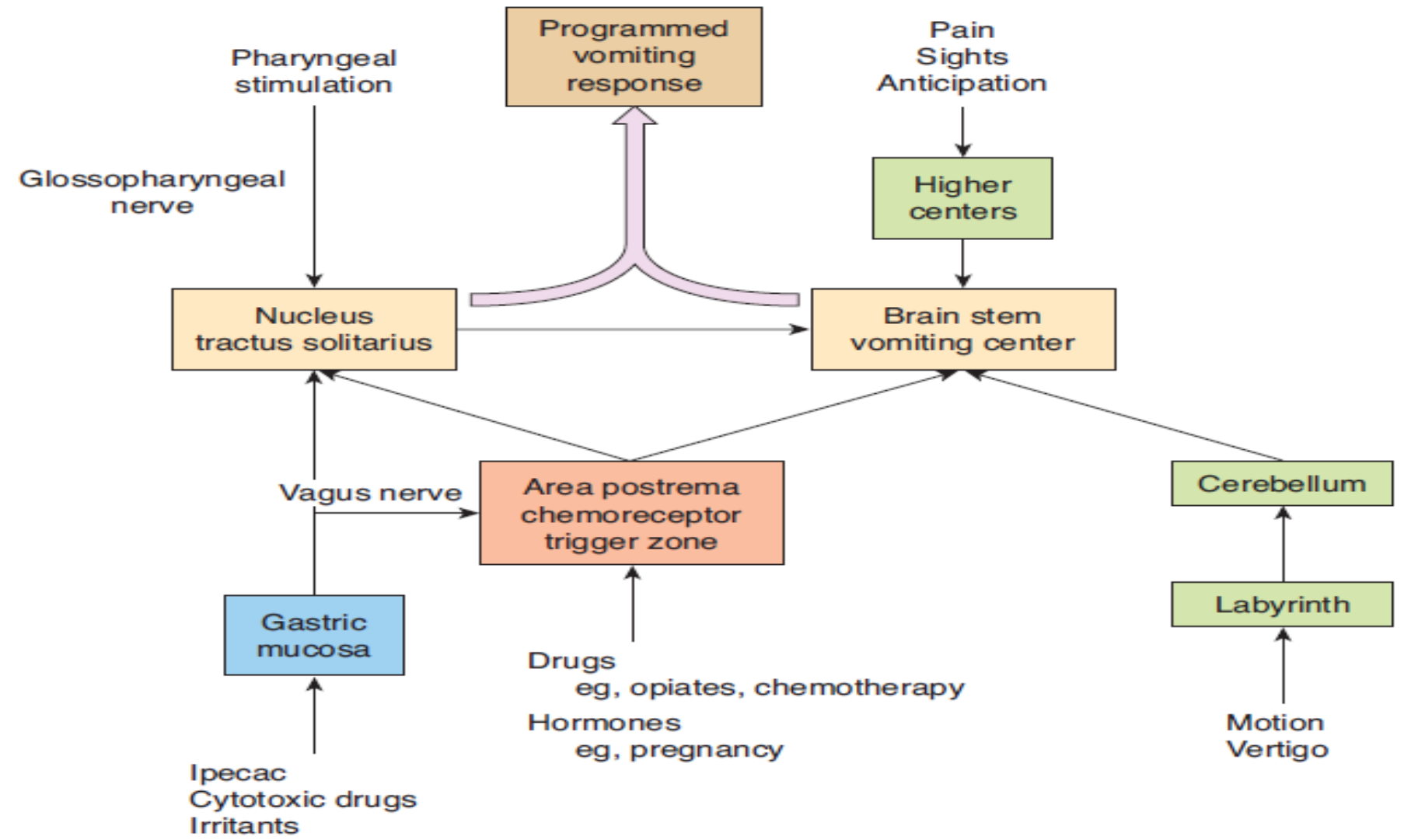
Clinical Correlates

Vomiting

- Starts with salivation and the sensation of nausea
- Reverse peristalsis empty materials from the upper part of the small intestine into the stomach
- The glottis closes, preventing aspiration of vomitus into the trachea (*the breath is held in mid inspiration*)
- Muscles of the abdominal wall contract which increases intra-abdominal pressure
- The lower esophageal sphincter and the esophagus relax, and the gastric contents are ejected

Vomiting

- **Vomiting center** is located in the reticular formation of the medulla
- Different factors can stimulate the vomiting center:
 - Pharyngeal irritation → GPN → NTS → Medulla
 - Gastric irritation → Vagus → NTS + Area postrema → medulla
 - Vertigo → cochlear labyrinth → vestibular nuclei → medulla
 - Pain/sights/smell/anticipation → diencephalon/limbic system → medulla



Neural pathways involved in vomiting

Vomiting

- Area postrema:
 - Located at the caudal end of the floor of the 4th ventricle
 - Relatively permeable part of the blood-brain barrier
 - Contains arterioles, sinusoids, and a few unipolar neurons
 - Chemoreceptor trigger zone → trigger vomiting in response to circulating emetic substances (*digitalis glycosides, apomorphine...*)
 - Lesions have little effect on vomiting caused by GI irritation/motion sickness
 - However, it abolish vomiting caused by circulating emetic substances (+ uremia and radiation sickness)

Vomiting

- Chemoreceptors triggering vomiting:
 - Serotonin (5-HT₃) → Small intestine
 - Dopamine (D₂) + 5-HT₃ → Area postrema + NTS
- **NB** → Antiemetic agents
 - 5-HT₃ antagonists → Ondansetron,
 - D₂ antagonists → Chlorpromazine + haloperidol...
 - Corticosteroids and Cannabinoids
 - Benzodiazepines (*reduce the anxiety associated with chemotherapy*)

Constipation

- Pathological decrease in bowel movements
- Difficulty in eliminating bowel movements from the body
 - Reflect changes in motility
 - Alterations in the balance between secretion and absorption in the colon
 - The only symptoms caused by constipation are slight anorexia and mild abdominal discomfort and distention

Diarrhea

- Characterized by excessive fluid excretion in the feces
 - Causes:
 1. Enterotoxin → produced by infecting bacteria (*cholera...*)
 - Release of enterotoxin stimulates active NaCl transport into intestinal lumen followed by the osmotic movement of water
 2. Damage of intestinal mucosa (*celiac sprue...*)
 - Intestinal mucosa of susceptible people is damaged by eating foods that contain gluten (proteins from grains such as wheat)
 3. Deficiency of Lactase (*Lactose intolerance*)
 - Undigested lactose increases osmolarity of intestinal lumen content → diarrhea

Appendicitis

- Acute inflammation of the vermiform appendix
- Commonly detected in its later stage by pain in the lower right quadrant of the abdomen
 - Pain and cramps in the area between the right hip bone and the navel, fever, nausea and vomiting, constipation, and diarrhea.
- If the appendix ruptures, infection can spread throughout the surrounding body cavity, causing **peritonitis** (*inflammation of the peritoneum*)
 - Treated by surgical removal of the inflamed appendix (appendectomy)